

HAEMOSTASIS

Hemostasis is the arrest or stoppage of bleeding. When a blood vessel gets damaged or injured, the injury initiates a series of reactions which causes hemostasis. Hemostasis occurs in 3 stages –

1. Vasoconstriction
2. Platelet plug formation
3. Coagulation of blood

1. Vasoconstriction

Immediately after injury, the blood vessels constrict and reduces the loss of blood from the injured tissue. Usually, arteries and small arterioles constrict. When the blood vessels are injured or cut, the endothelium is damaged and collagen is exposed. The platelets adhere to this collagen and gets activated. The activated platelets secrete serotonin and other vasoconstrictors substances which causes the vasoconstriction. The adherence of platelets to the collagen is accelerated by von Willebrand factor.

2. Platelet plug formation

When platelet gets adhered to the collagen and gets activated, it secretes adenosine diphosphate (ADP) and thromboxane A_2 which attracts more platelets to the damaged tissue and activates them. These platelets aggregate and form a temporary platelet plug which closes the ruptured endothelium and reduces the further blood loss. This temporary platelet plug is also known as hemostatic plug. Platelet aggregation is accelerated by platelet-activating factor (PAF).

3. Coagulation of blood

During this process, the fibrinogen gets converted into fibrin and forms a mesh like appearance and gets attached to the platelet plug which block the ruptured part of the blood vessels and prevent the loss of blood completely.

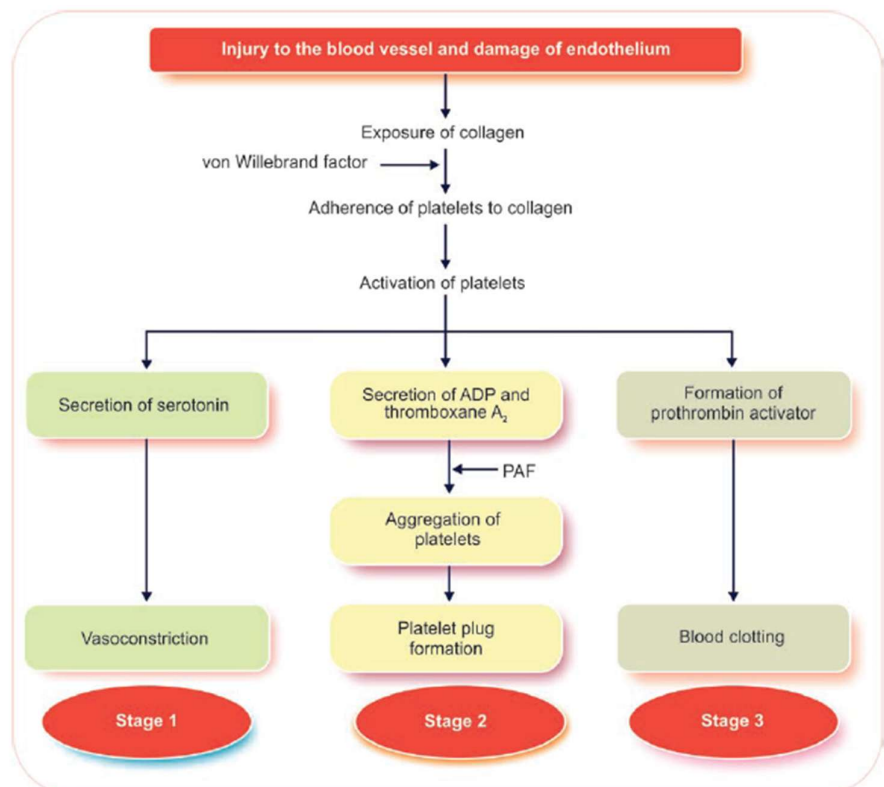


FIGURE 19.1: States of hemostasis. ADP = Adenosine diphosphate; PAF = Platelet-activating factor.